# Predicting NBA Shot Outcomes Using Location, Shot Context, and Machine Learning

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**Figure 1: Golden State Warriors at the 2022 NBA Final Championship.**

## Abstract

In today's game of basketball, shot selection is quite crucial. Knowing what makes a shot successful may be very helpful for players, coaches, and analysts. This effort seeks to forecast whether a shot is made or missed based on over four million NBA shot attempts for several seasons. We perform an exploratory data analysis to examine the shooting percentages by shot type and court zone. We compare several machine learning models (Logistic Regression, Decision Trees, Random Forest, etc.). We assess model performance based on accuracy, F1 score, and ROC AUC.

Our results reveal that generally the Random Forest model performed the best with a ROC AUC of around 0.67. An examination of feature importance demonstrates that shot distance, shot type, and court location are among the most important determinants of shot success. We also analyze whether the inclusion of player identity enhances prediction accuracy, concluding that it does little to improve prediction accuracy. These results suggest that, at a league level, the context and position of a shot are more predictive of the outcome of the shot than the specific player taking the shot.

## CCS Concepts

• **Applied computing** → **Sports analytics**; • **Computing methodologies** → *Machine learning*; • **Information systems** → *Data mining*.

## Keywords

NBA, Sports Analytics, Machine Learning, Shot Prediction, Random Forest, Basketball Analytics

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# 1 Introduction

Basketball is one of the most data-driven sports in the world. Today's tracking technologies capture masses of data on every shot attempt – where it came from on the floor, the type of shot, the game circumstances and who took the shot. These data provide an opportunity to better understand the factors that determine whether a shot is made or missed.

Basketball strategy is mostly based on shot selection. Coaches and analysts generally judge offensive efficiency by where shots are taken and what shot types are attempted. Previous research has demonstrated that shooting success can be considerably influenced by factors such as shot distance, court location, and player skill [2]. It is a difficult challenge predicting the outcome of an individual shot because of the large number of variables that come into play during a game.

This project examines the ability of machine learning models to predict the outcome of NBA shots based on historical data at the shot level. We utilize a big dataset of millions of NBA shots taken throughout several seasons. Each observation contains information on shot distance, shot position, shot type, court zone, game context, and player information.

The analysis aims at three key purposes. We start by looking at patterns in shooting efficiency for various shot types and zones on the floor. The second is comparing several machine learning algorithms to forecast whether a shot will be made or missed. Finally, we assess which features contribute most to model performance and evaluate if player identity has considerable predictive value.

We evaluate the performance of four prediction methods: majority baseline, Logistic Regression, Decision Tree, and Random Forest. Overall, the Random Forest model performed the best with an accuracy of 63.4% and a ROC-AUC score of 0.67. The feature relevance analysis indicates shot distance, shot location and shot context are better indicators than the identification of the individual player.

The results show that despite the high randomness of the shot outcomes, machine learning models can extract meaningful patterns that outperform basic baselines. The findings provide insight into shot selection in basketball and the application of predictive analytics in sport.

# 2 Related Work

Recent advances in sports analytics have resulted in a growing interest in the prediction of basketball performance using statistical and machine learning techniques. Early basketball analytics were largely descriptive data like FG %, shot charts, player efficiency stats, etc. With the availability of larger datasets, prediction models were used to examine player and team performance.

A number of studies have used the shot location data to analyze shooting efficiency on different areas of the court. In all this research, it was observed that shots taken closer to the basket often had a higher success rate, whereas long-distance shots carry greater risk but have higher scoring rewards. Another study has used contextual elements such as game time, defensive pressure, and player tracking data to improve prediction accuracy [1].

Also, machine learning techniques such as Logistic Regression, Decision Trees, Random Forests, Gradient Boosting, and Neural Networks have been widely used in basketball prediction problems [3]. Much of this research relies on specialized datasets with player tracking or enhanced optical tracking data. Instead, our work employs readily available shot level variables that are directly available from the public NBA shot datasets.

Our work is different from much of the previous literature in that we compare classical machine learning models as well as exploratory basketball analytics in the same setting. We also explore the value of features and whether the identity of the player provides additional useful information beyond shot location and game context.

# 3 Dataset and Methods

## 3.1 Dataset

For this study, we used a collection of NBA shot-level data over 21 seasons. Each row of the data represents a single shot attempt, and includes information on the shot location, shot type, game context, and shooter. The response variable is an indicator variable for whether the shot was made, `SHOT_MADE`.

We selected several factors for modeling. Spatial information includes shot distance and x and y coordinates of the shot position on the court. Contextual variables were quarter, minutes remaining, and seconds remaining. Other category variables were shot type, action type, court zone, and player position.

Because the original dataset included millions of observations, a random sample of 500,000 shot attempts was used to train and evaluate the models. Before modeling, missing values for important variables such as position and player information were imputed. The generated dataset offered a sufficiently big sample size, while preserving adequate processing performance.

We explored shooting efficiency and shot volume in different regions of the court to better understand shooting behavior before building models.

## 3.2 Data Preprocessing

Before developing prediction models, several preprocessing steps were completed. We removed missing values in critical variables to avoid noise during training. One-hot encoding was used to transform categorical data, including shot kind, court zone, action type, and player identity, into numerical representations.

Numerical variables such as shot distance, court locations, game clock information, and quarter were returned to their original formats. The target variable was whether the shot went in or missed.

We split the dataset into training and testing subsets with an 80/20 split to evaluate model performance. The model was then evaluated on the held-out test set.

## 3.3 Models

We used three different classification models to predict whether an NBA shot was made or missed.

The first model was Logistic Regression. We picked it as a baseline since it is simple, fast to train, and often used for binary prediction applications. This offered us a benchmark before we tried more complicated models.

After that, we trained a Decision Tree model. In contrast to Logistic Regression, a Decision Tree is able to learn more complex

correlations between factors such as shot distance, court location, and game circumstances. And it provides an intuitive insight into how the forecasts are created.

Finally, we trained a random forest model that merges multiple decision trees into one ensemble model. In our studies, Random Forest scored the highest ROC-AUC score and had best overall performance. Therefore, most of our analysis is based on the results of the Random Forest.

We compared the models on the test set by prediction accuracy and ROC-AUC scores. Furthermore, we also conducted feature importance analysis to better identify the most contributing aspects to shot success.

## 4 Exploratory Data Analysis

We performed exploratory research to determine patterns in shooting efficiency and shot selection across the floor. The study of these patterns provides significant context for understanding the machine learning models developed later in the project.

Figure 2 plots the two-point field goal % against the three-point field goal %. In general, two-point shots are far more successful than three-point shots because three-point shots are further and more difficult to make. However, they also earn you more points.

Figure 3 investigates shooting efficiency across various court zones. The limited region has the highest field goal percentage (over 60%) and the long distance zones, such as above the break three-pointers and corner threes, are much lower. Backcourt shots are extremely rare and the least efficient.

Figure 4 shows the total shot volume by court zone. Even if the restricted area is still the most attacked region, we see a lot of tries from the mid-range and the three-point above the break. This suggests that player decision making involves a trade-off between efficiency and scoring opportunity.

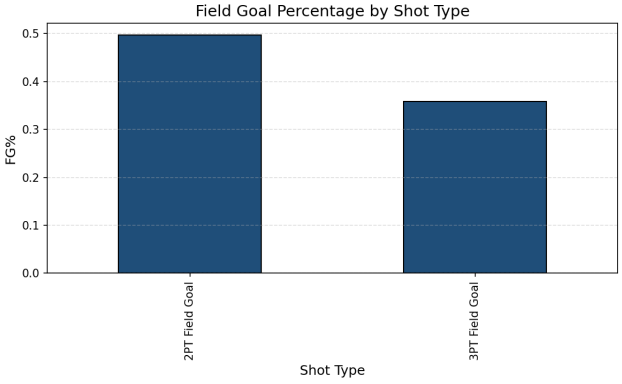



**Figure 2: Field goal percentage by shot type.**

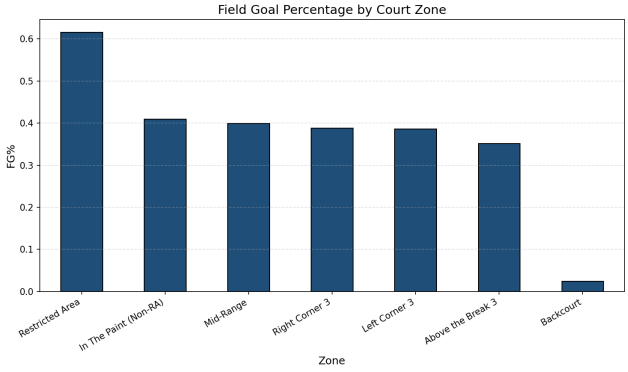



**Figure 3: Field goal percentage by court zone.**

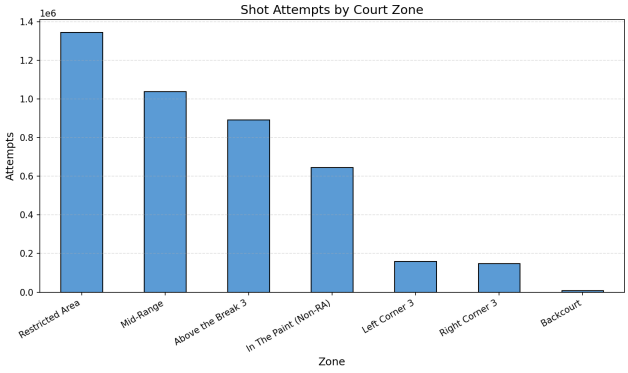



**Figure 4: Shot attempts by court zone.**

## 5 Model Results

Following the exploratory study, different classification models were trained and assessed to predict whether an NBA shot would be made or missed. Among the models examined, Random Forest consistently showed the best performance and was therefore picked as the final model for detailed study.

The Random Forest classifier achieved a test set accuracy of approximately 63.4% and a ROC-AUC score of about 0.67. While the outcome of a shot is intrinsically a random event, our results demonstrate that meaningful patterns can still be retrieved from the shot-level information.

The fifteen most significant features as determined by the Random Forest model are shown in Figure 5. Shot distance was the most influential variable, followed by other aspects of shot type and court location. Variables that described where a shot was taken regularly ranked near the top. It shows that knowledge about position is very important to make a shot successful.

Interestingly, player identity elements added relatively little compared to shot context data. This suggests that, at least at the league level, shooting trends are more a function of shot quality than individual shooter.

The confusion matrix for the final Random Forest model is given in Figure 6. The algorithm is quite good at identifying the missed

shots, properly classifying a lot of unsuccessful efforts. As expected, predicting made shots is more difficult since successful shot outcomes are dependent on several criteria not fully represented in the dataset.

To further analyze the classification performance, we further investigated the ROC curve displayed in Figure 7. The curve is above the diagonal baseline at all times, which indicates that the model is better than random guessing at various decision criteria.

The ROC-AUC score is around 0.67, which means the model has moderate capacity to predict. The model is far from perfect, but it shows that shot location, shot type, and game context convey useful information about the probability of a shot to be successful.

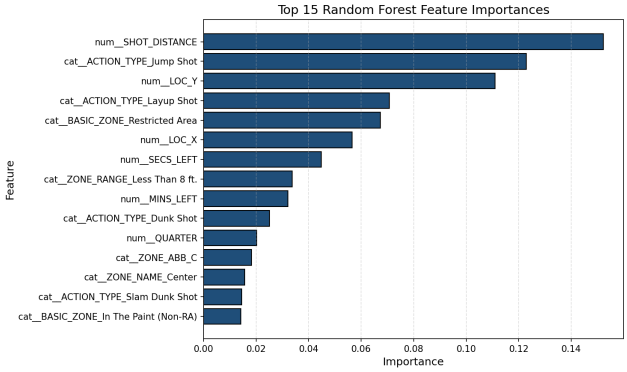



**Figure 5: Top 15 Random Forest feature importances.**

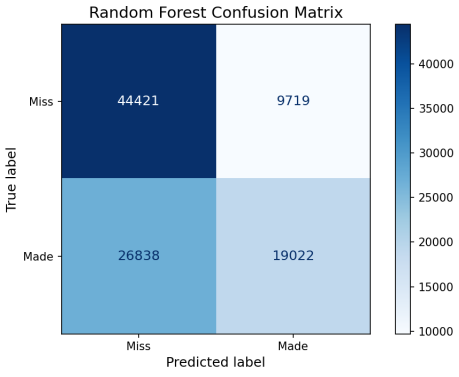



**Figure 6: Random Forest confusion matrix.**

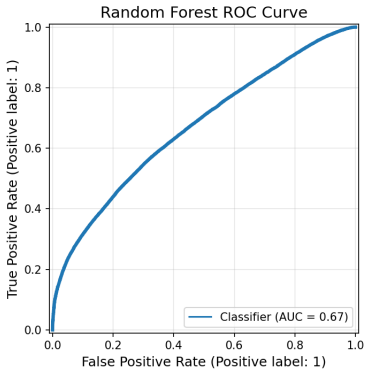



**Figure 7: Random Forest ROC curve.**

## 6 Limitations

Several limitations should be taken into account when evaluating the results of this study.

First, the dataset only contains information available in NBA shot logs. Important background such as defensive pressure, shot contest, weariness, and team strategy are not available. Many of the inaccuracies in the prediction by the model are likely due to these missing factors.

Second, the approach pools shots across seasons and players into a single prediction assignment. This improves the sample size but can obfuscate differences between players' shooting habits and the offensive schemes of other teams.

Third, only standard machine learning models are considered. More advanced techniques such as gradient boosting algorithms, deep neural networks, or sequence-based models may be better at prediction.

Finally, the model performance was assessed on historical data only. Further work could explore the generalization of the model to new seasons or playoff games.

## 7 Conclusion

In this project, we explored whether the result of an NBA shot can be predicted using shot location, shoot context, game information and player identifiers. We initially examined shooting tendencies across different areas of the court and developed many machine learning models to predict whether a shot was successful or missed.

The exploratory research revealed considerable disparities in shooting efficiency in different areas of the court. Those taken close to the basket were most successful, while those from far range were often less efficient. In some of the lower-efficiency locations, there were a lot of high shots, which shows the tradeoff between how hard the shots are and how valuable they are for scoring in modern basketball.

Overall, the Random Forest model performed best of the models examined. Among the predictors, shot distance and court location were shown to be the most informative, with player identification contributing less than anticipated. This shows that, league-wide, the context of the shot is often more relevant than the player in forecasting outcomes of shots.

The results suggest that the publicly accessible NBA shot data do contain relevant information that can be used for prediction, even though the model is far from ideal. Future studies might include tracking data, defensive information and player movement data to increase performance and better portray the complexity of real game circumstances.

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